05 Connecting the three evidence streams

Four boundary-specific syntheses. Existing mechanisms are not automatically validated in the material-flow domain.

Boundary	S–U: requirement / gap	S–DT: domain function	DT–U: candidate mechanism	Synthesis / proposed addition
S1 → S2	U03 / U05 / U06 Attribute and MI variability may become point inventory.	B05 / B03 BIM–GIS quantity mapping; alternatives can be averaged.	004 / 016 Conditional or covariance representations preserve selected dependencies.	Keep material alternatives, shared dependencies and uncertainty through the quantity interface.
S2 → S3	U11 / U12 / U18 Lifetime and stock assumptions change flow timing.	B03 Layer masses + service lives produce annual outflow CSV.	004 / 016 Sequential inference and measured feedback can update represented states.	Link inventory realizations to conditioned lifetime models; revise with new event evidence.
S3 → S4	Selected-set gap No assessed direct uncertainty chain from outflow to recoverable quality.	B04 / B11 Reuse/disassembly information; not a verified probabilistic recovery-state transition.	018 (analogy) Consistency / prognostics mechanisms; own covariance handoff remains limited.	Specify condition, damage, contamination and routing interfaces. Domain data and validation are required.
S4 → S5	U19 Shared recovery inputs; joint draws and dependence insufficiently documented.	Not established here Focused domain bridge set does not establish a direct S4–S5 DT architecture.	016 (analogy) Covariance/cross-covariance transport for represented performance quantities.	Define joint recovery-to-impact input realizations and burden/credit mappings; validate in the target domain.

B aliases identify domain/enabling papers; numeric 004/016/018 labels identify DTUM studies. Full identifiers, DOIs and rationales are in the CSVs.

Dashed arrows are reviewer synthesis/proposals. Missing direct evidence is retained; no complete deployed S1–S5 system is claimed.